



Problem-Solution Fit

Date	15 July 2026
Team ID	xxxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

This document maps out the target student population, their psychological limitations, existing alternative tools, behavioral patterns, and how our **Emotion Detection & Learning Support Engine** directly establishes a proper market and product fit..

1. CUSTOMER SEGMENT(S) [CS] Technical university students (e.g., Computer Science, Engineering departments) who balance heavily demanding schedules, rigid coding laboratory projects, and continuous testing regimes.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] ☐ Limited financial resources for private human tutoring subscriptions. ☐ Relies on standard student laptops; requires a lightweight web client execution environment that does not strain their physical hardware during inference processing.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Option A: Generic LLMs (ChatGPT / Standard Chatbots): • <i>Pros:</i> Instantly available. • <i>Cons:</i> Lack specialized deep-learning sentiment categorization; provide overwhelming, non-contextual walls of text rather than managing the user's emotional state.
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		Option B: University Counseling / TA Office Hours: • <i>Pros:</i> Deeply personalized and highly human. • <i>Cons:</i> Completely unavailable during late-night study sessions; presents a social barrier for students hesitant to report struggle openly.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] ☐ Sudden cognitive roadblocks, severe programming logic bugs, and high anxiety phases right before academic evaluations. ☐ Frequency: Happens multiple times per week, peaking intensely late at night during peak project submission windows.	9. PROBLEM ROOT / CAUSE [RC] Traditional computer science curricula focus exclusively on rigid syntax rules and grading outputs, leaving out immediate, empathetic micro-guidance when a student faces a steep learning curve or mental block.	7. BEHAVIOR + ITS INTENSITY [BE] Venting out academic stress in lengthy text blocks, erratic late-night typing loops, and high-intensity procrastination cycles triggered by emotional burnout.
3. TRIGGERS TO ACT [TR] Compilation exceptions that won't resolve, consecutive failed automated test scripts, impending project assignment deadlines, or mounting stress before a major exam.	10. YOUR SOLUTION [SL] An integrated Emotion Detection & Learning Support Engine that detects real-time sentiment metrics (Anxiety, Frustration, Sadness) using multi-model PyTorch classifiers (BERT/BiLSTM) and passes those states to the Gemini API to formulate direct, stress-reducing study tactics.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Venting across informal peer Discord servers, technical subreddits, or scouring Stack Overflow blindly.
4. EMOTIONS BEFORE / AFTER [EM]		OFFLINE Isolating in university dorm rooms, staying silently frustrated during lab

□ Before: Overwhelmed, highly frustrated, isolated, and anxious about potential academic failure.

□ After: Supported, clear-headed, motivated, and relieved by having a structured, bite-sized study path forward.

assignments, or avoiding instructor contact.